Towards a Numerical Digital Twin of the Solar Gravitational Lens

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1 The SGL Opportunity

The Sun acts as a gravitational lens, curving photon trajectories through spacetime. For a perfectly aligned source, Sun, and observer, the lensed light forms an Einstein ring.

The Solar Gravitational Lens (SGL) offers potentially extraordinary optical amplification and angular resolution. Unlike a conventional telescope, the lens itself is provided naturally by the Sun.

> **Could the Solar Gravitational Lens be developed into a practical astronomical observatory?**

An SGL mission could potentially enable observations at resolutions inaccessible to conventional telescopes. But realizing that capability requires more than the existence of the lens itself.

The central challenge is turning the theoretical SGL into a **physically and computationally understood observation system**: given an astronomical source and an observer geometry, what photons reach the observer, from what directions, and how does the result change as the geometry changes?

Figures: SGL mission geometry showing source, Sun, spacecraft, and representative lensed photon trajectories.

2 The Computational Gap

An SGL mission is not simply a matter of placing a spacecraft beyond the Sun's focal region and collecting light. The observed signal depends on source–Sun–spacecraft geometry and on the detailed propagation of photons through the gravitational field.

The first computational requirement is therefore a **numerical forward model**:

$$\boxed{\text{source + observer geometry} \longrightarrow \text{GR photon propagation} \longrightarrow \text{observable at the observer}}$$

The goal is not simply to visualize individual photon trajectories. The goal is to turn those trajectories into measurable quantities and, ultimately, a predicted photon distribution at the observer.

The present work establishes this forward-model core in Schwarzschild spacetime and develops it from an idealized Einstein-ring calculation toward explicit 2D photon sampling.

3 The Core Technology: Penrose + SGL

The Solar Gravitational Lens model is built on top of *Penrose*, a generalized general-relativistic simulation engine developed to numerically model particle and photon motion in curved spacetime.

Penrose is designed as a physics engine rather than an SGL-specific simulator. Its core abstraction is the propagation of particles and photons through a specified spacetime using numerical geodesic integration.

The Schwarzschild photon propagation used here is therefore not a new component written specifically for the SGL. It is an existing capability of the generalized Penrose engine, with the SGL model providing the geometry, initial conditions, sampling, and observation layer around it.

Conceptually,

$$\boxed{\text{Penrose} \rightarrow \text{GR photon propagation} \rightarrow \text{SGL forward model}}$$

The GR engine remains independent of the physical scenario. This makes the SGL one application of a broader computational GR foundation rather than a standalone ray-tracing implementation.

4 From Photon Propagation to the Einstein Ring

The core numerical calculation is a null-geodesic boundary-value problem.

A photon trajectory is obtained by integrating

$$\frac{d^2x^\mu}{d\lambda^2} + \Gamma^\mu_{\alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda} = 0$$

in Schwarzschild spacetime using numerical geodesic integration.

For the aligned 1D model, a family of trial rays is parameterized by b. Each trajectory crosses the observer plane at some residual position $u(b)$. The calculation searches for the observer-hitting trajectory,

$$u(b_\star) = 0,$$

and then extracts its incoming angular position,

$$\boxed{\theta_E = \arctan\rho}$$

where ρ is the observer-frame gnomonic angular coordinate.

For the perfectly aligned source–Sun–observer configuration, the geometry has rotational symmetry about the optical axis. The observer-hitting trajectory therefore represents an azimuthally degenerate family of equivalent rays, forming an Einstein ring.

Thus the ring is reconstructed from the observer-hit angular coordinate rather than measured from the pixels of the rendered image.

This provides the first structural result of the forward model: **the aligned SGL geometry produces the expected Einstein-ring structure through GR photon propagation and observer mapping**.

Figures: Simulated Einstein ring and θ_E versus source distance with the weak-field analytical comparison.

5 Finite Source Distance: From an Idealization to a Parameter Study

Much analytical treatment of the SGL begins with a source at effectively infinite distance, so the incoming photon field can be approximated as parallel. Turyshev and Toth explicitly identify finite source distance as an important extension of this idealization [1].

The numerical model therefore asks a more general question:

> **How does the observer's Einstein angle change as the source moves from finite distance toward the far-field regime?**

For a finite source distance S, the incoming photons are no longer exactly parallel. Changing S changes the photon initial conditions and the source–lens–observer geometry, and therefore changes the trajectory that reaches the observer.

The numerical experiment evaluates

$$\theta_E = \theta_E(S)$$

over a range of source distances.

The computed observer-hit angle varies systematically with S and approaches the parallel-beam reference as the source becomes more distant.

Figure: Gnomonic radius ρ versus source distance.

The significance of this experiment is that the forward model is no longer producing a single idealized picture. It is beginning to act as a **parameterized scientific model** for how an SGL observable depends on source geometry.

Important scope: the current simulations use geometrized units with $r_s = 1$. The "infinite" configuration is a finite parallel-beam stand-in, not a numerical realization of $S = \infty$, and the experiment should not be interpreted as a solar-scale mission simulation.

Reference: S. G. Turyshev and V. T. Toth, "Imaging extended sources with the solar gravitational lens," *Physical Review D* 100, 084018 (2019).

6 From Symmetry to 2D Photon Sampling

The symmetric model is computationally efficient, but it relies on a strong idealization: perfect on-axis symmetry.

A more general photon-field calculation requires sampling many independent photon trajectories explicitly.

The model therefore moves to a 2D launch plane, where photon initial conditions are sampled over two transverse coordinates and propagated through the same GR engine.

The resulting pipeline is:

$$\boxed{\begin{array}{c}\text{2D photon sampling}\\ \downarrow \\ \text{geodesic integration}\\ \downarrow \\ \text{observer-plane intersections}\\ \downarrow \\ \text{photon distribution}\end{array}}$$

This is an important transition in the project:

$$\boxed{\text{symmetry-assisted calculation} \longrightarrow \text{explicit photon-field sampling}}$$

The same Penrose GR integrator remains underneath the new sampling strategy.

For the aligned case, symmetry can still be used in the final visualization; the purpose of the 2D formulation is to remove that dependence from the underlying photon search and to provide a route to more general configurations, including off-axis observer geometries.

Figure: 2D simulated Einstein ring.

7 Scaling the Calculation

Explicit 2D photon integration increases the computational cost rapidly with the number of trajectories.

However, the structure of the problem is favorable for parallel computation: individual photon geodesics evolve independently in a fixed background spacetime.

At the geodesic level,

$$\text{photon } i \quad \text{does not depend on} \quad \text{photon } j.$$

CPU parallelization has therefore already been implemented, allowing independent trajectories to be distributed across CPU threads while retaining the existing scientific implementation as a reference path.

The same structure makes the problem a strong candidate for GPU acceleration.

The computational progression is therefore:

$$\boxed{\text{Single trajectory} \rightarrow \text{Parallel photon ensemble} \rightarrow \text{GPU-accelerated photon field}}$$

The objective is not acceleration for its own sake. The objective is to make larger photon ensembles, higher-resolution photon fields, and more general SGL calculations computationally feasible.

Figure: Speedup versus CPU threads.

8 What the Current Model Establishes

The present system should be viewed as a **numerical foundation**, not as a mission-grade SGL observatory.

Established capability:

- Schwarzschild null-geodesic propagation;
- aligned Einstein-ring modelling;
- extraction of an observer-hit angular observable;
- finite source-distance characterization;
- explicit 2D photon sampling;
- CPU-parallel photon propagation.

Current boundaries:

- geometric optics rather than wave-optical diffraction;
- point-source and point-observer assumptions;
- no detector, photometry, or spacecraft dynamics;
- Schwarzschild vacuum geometry rather than a full solar environment;
- normalized geometrized units rather than solar-scale execution.

These boundaries are deliberate. The present objective is to establish and test the computational physics core before adding the layers required for mission-level modelling.

9 Towards a Numerical Digital Twin

The long-term objective is an end-to-end numerical representation of an SGL observatory.

The conceptual pipeline is:

Astronomical source
$\downarrow$
GR photon propagation
$\downarrow$
Solar gravitational lens
$\downarrow$
Spacecraft trajectory
$\downarrow$
Detector
$\downarrow$
Image reconstruction

Current: numerical GR propagation, symmetric SGL modelling, finite source-distance characterization, explicit 2D sampling, and CPU parallelization.

Next: GPU acceleration, higher-resolution photon fields, and increasingly general source and observer geometries.

Long term: incorporate the physical and instrumental layers required to study an actual SGL observing system, including source structure, wave-optical effects, solar environment, spacecraft, detector, and image reconstruction.

The intended outcome is not simply a simulator that produces pictures. It is a computational platform for asking mission-level questions before a physical observatory exists.

Key Message

The Solar Gravitational Lens could provide an extraordinary new astronomical capability. The challenge is turning that physical possibility into a quantitatively understood observation system.

This work establishes the computational foundation for that effort: a generalized GR engine coupled to an SGL forward model, demonstrated on the aligned Einstein-ring configuration, extended to finite source distances and explicit 2D photon sampling, and structured for scalable photon-field simulation.